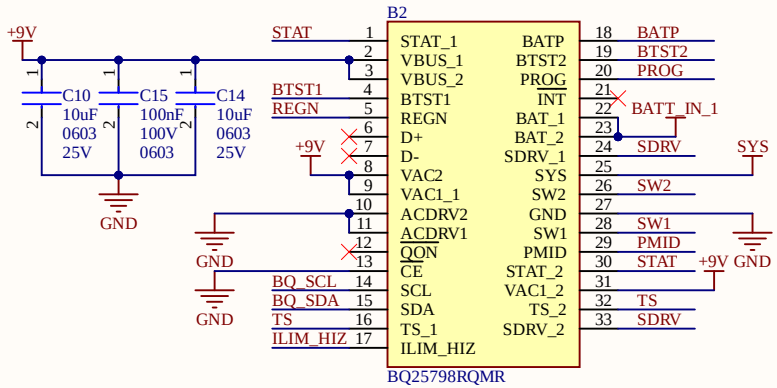
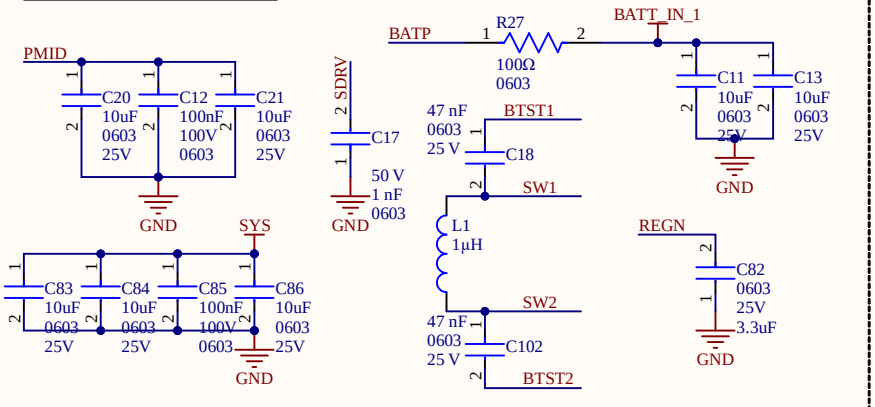
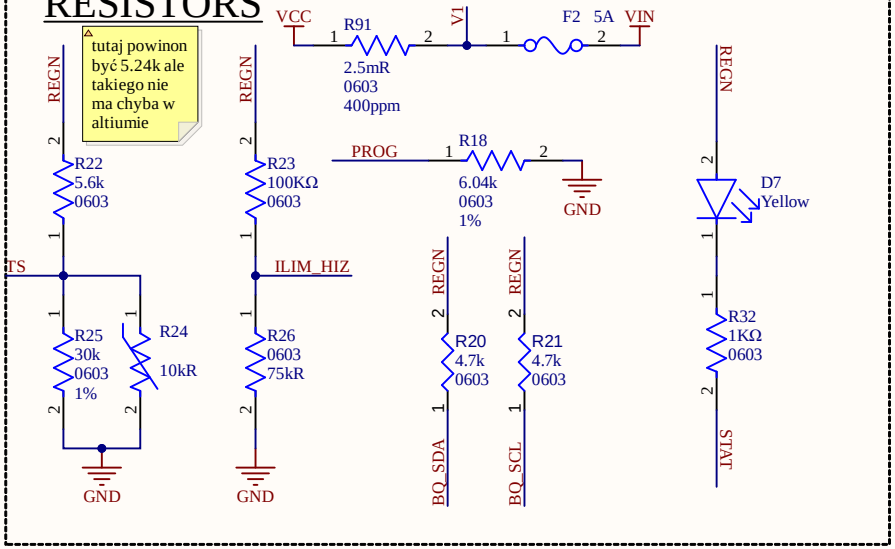
BATT_CHARGER



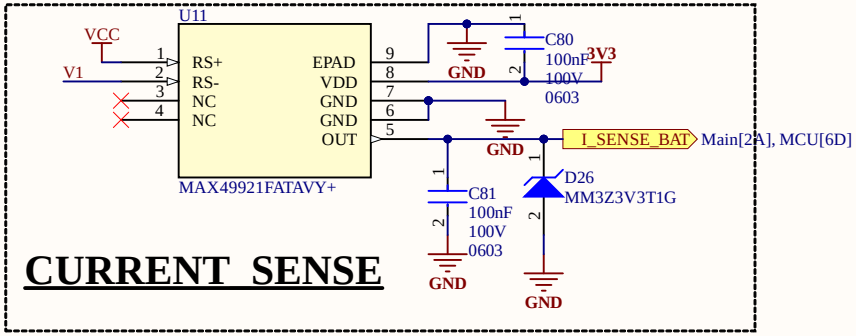
CAPACITORS



RESISTORS



CURRENT_SENSE



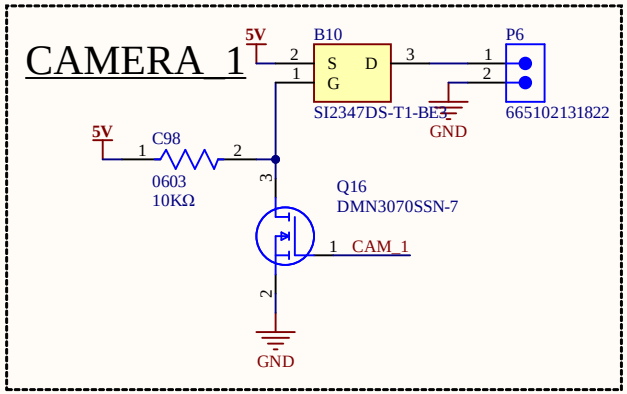
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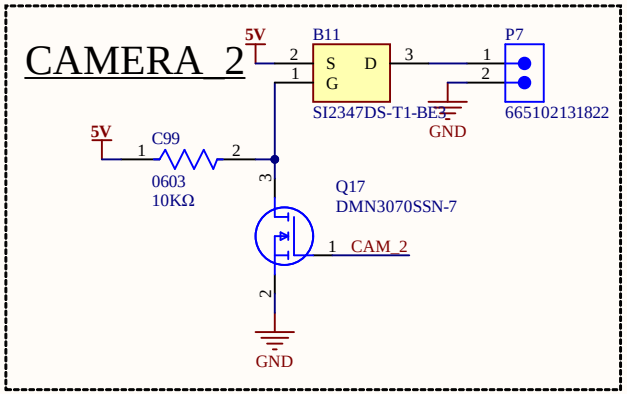
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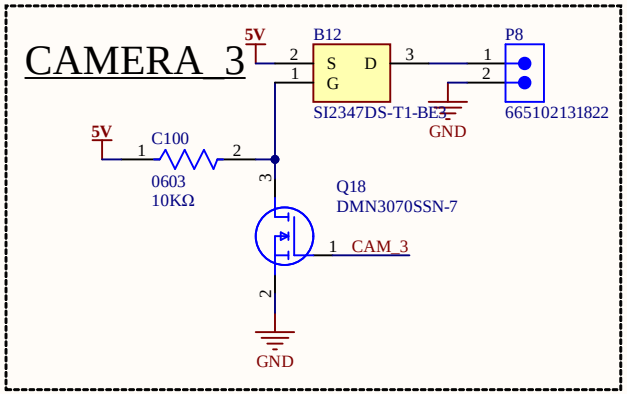
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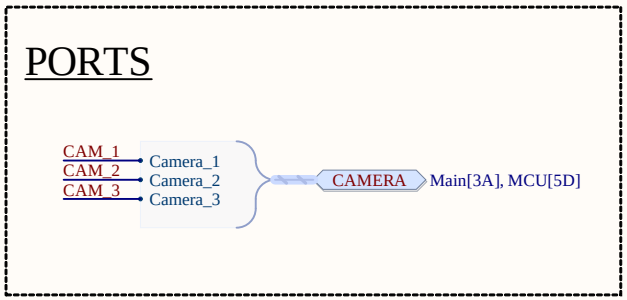
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Reviewers:









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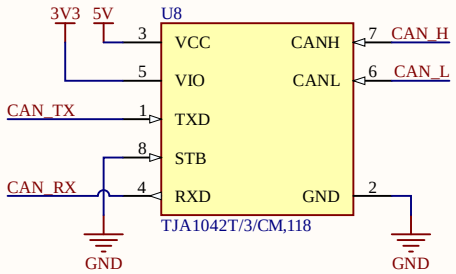
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Reviewers:

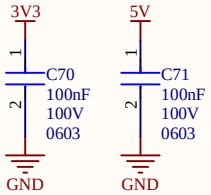
CAN

70mA w peaku

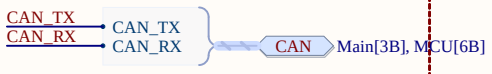
CAN TRANSCIEVER



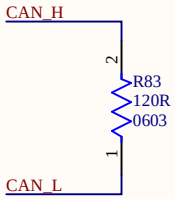
VCC & VIO Decoupling



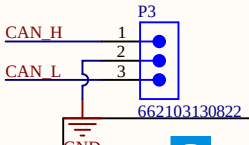
CAN



Termination



Connector



PWR IN SPACE

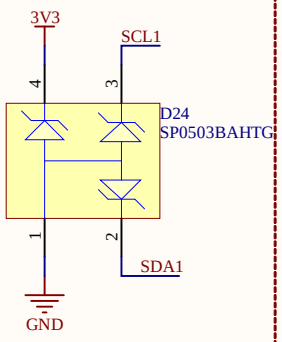
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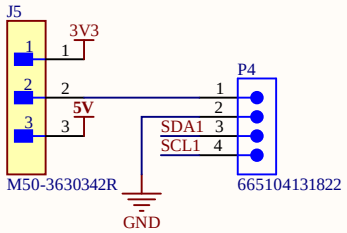
Description:

Reviewers:

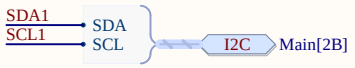
TVS_I2C



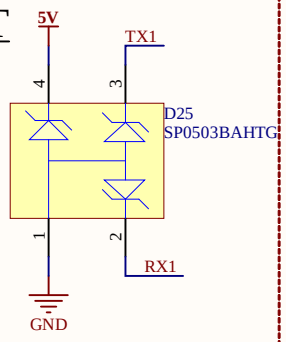
Connector



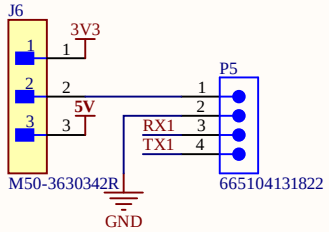
Ports



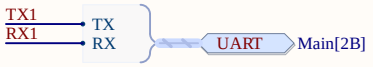
TVS_UART



Connector



Ports



PWR IN SPACE

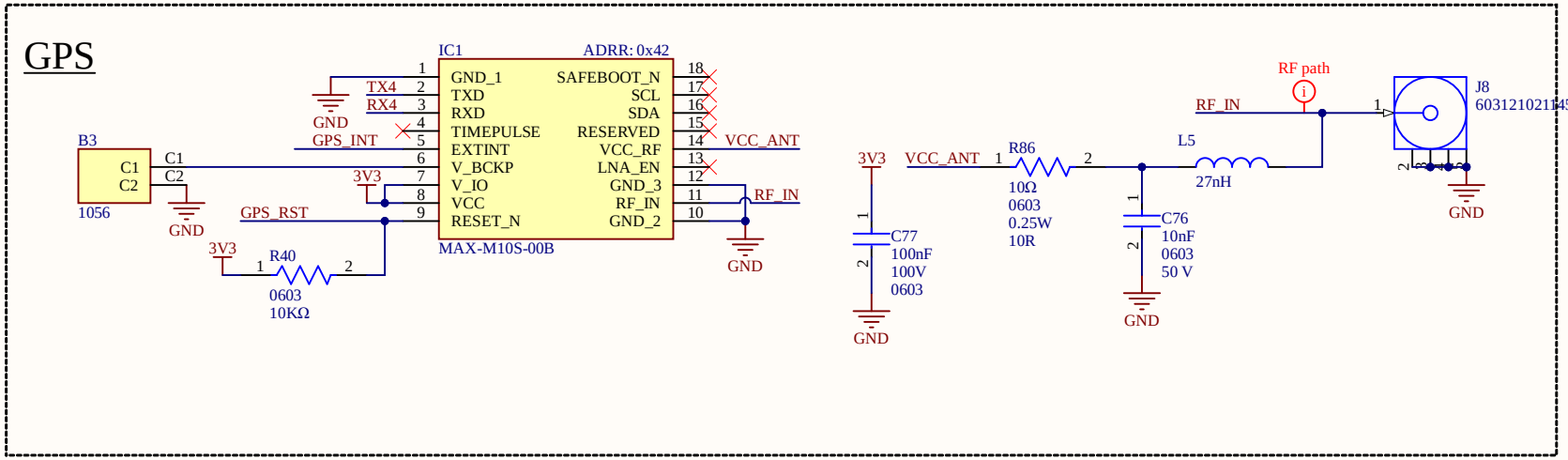
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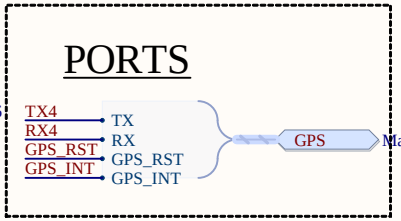
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Description:

Reviewers:

GPS

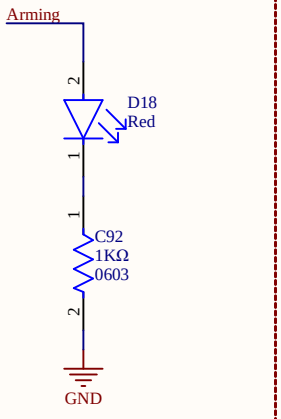




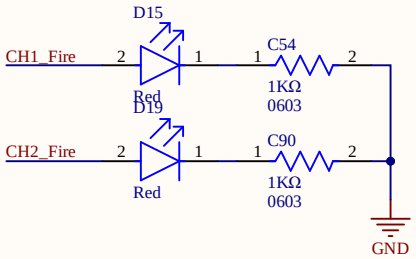
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	Sheet number: * / * ***Modified:*** 23.08.2026
Description: ***Reviewers:***	

Indicators

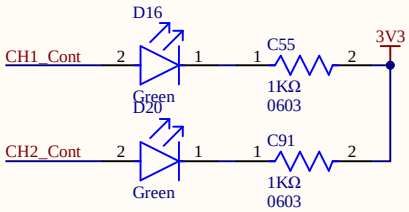
Arming Indicator



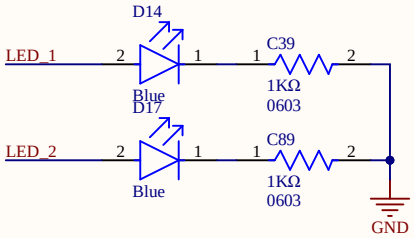
Pyro Fire Indicators



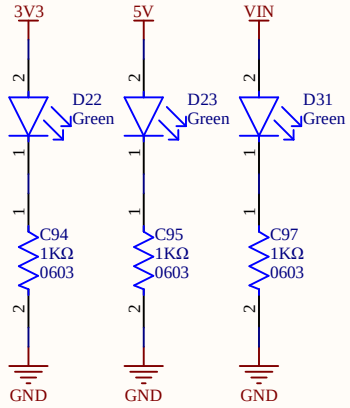
Pyro Continuity Indicators



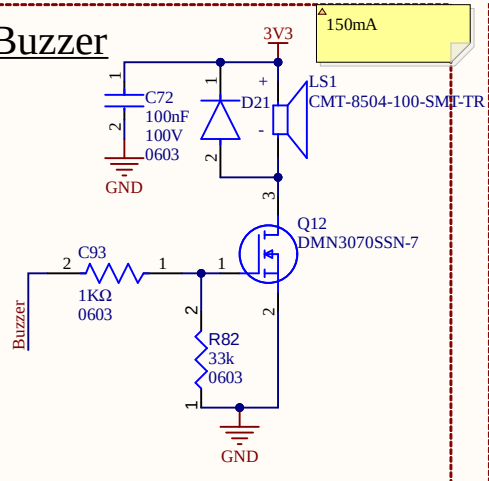
General use LEDs



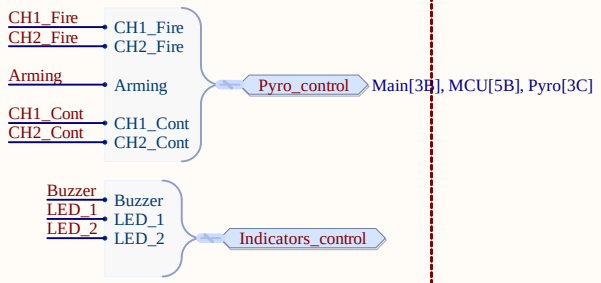
Power Indicators



Buzzer



Ports



PWR INSPACE

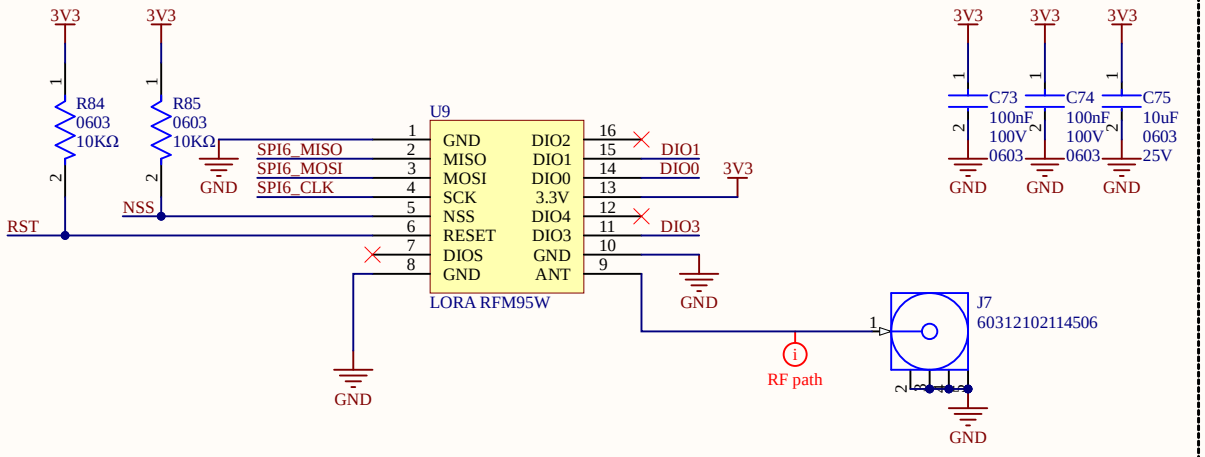
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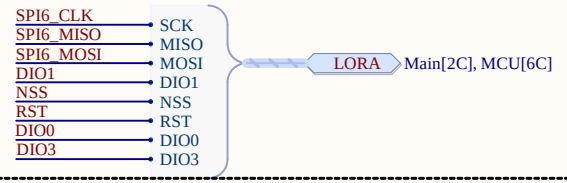
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Reviewers:

LORA



PORTS



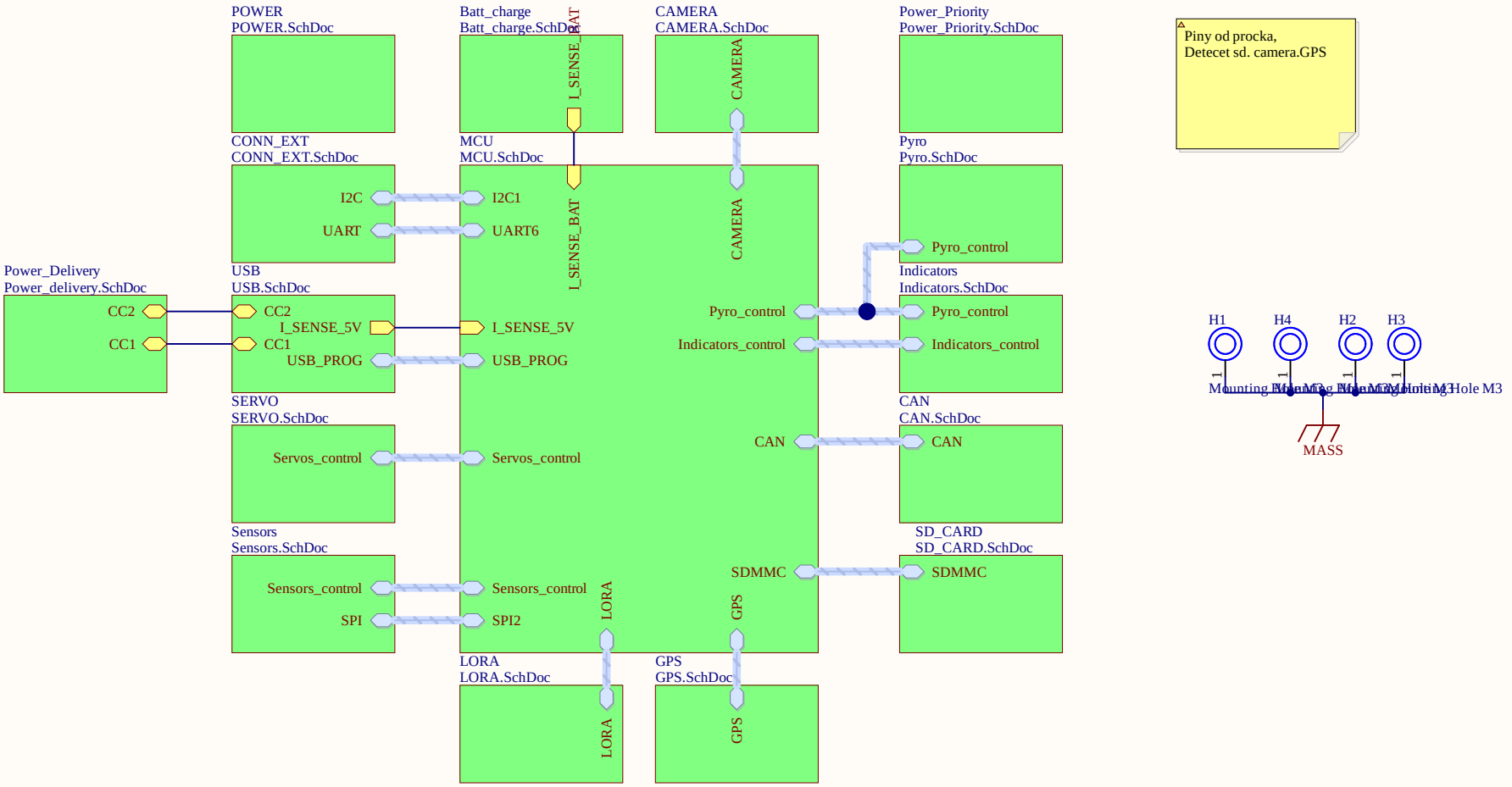
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Description: ***Reviewers:***	

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REVISION	DESCRIPTION	DATE	APPROVED

MCU

STM32H725VGT6

U4A — 1 of 2 — STM32H725VGT6

BOOT	94	BOOT0	
NRST	15	NRST	
CH3_PWM_MCU	37	PB0	
CH2_PWM_MCU	38	PB1	
	39	PB2	
SPI6_CLK	89	PB3	
SPI6_MISO	90	PB4	
SPI6_MOSI	91	PB5	
DIO1	92	PB6	
NSS	93	PB7	
CAN_RX	95	PB8	
CAN_TX	96	PB9	
SCL1	42	PB10	
SDA1	43	PB11	
	48	PB12	
SPI2_CLK	49	PB13	
SPI2_SDI	50	PB14	
SPI2_SDO	51	PB15	
	82	PD0	
DETECT_SD	83	PD1	
CMD_SD	84	PD2	
RST	85	PD3	
DIO0	86	PD4	
DIO3	87	PD5	
MAG_CS	52	PD8	
MAG_INT	53	PD9	
BARO_CS	54	PD10	
IMU_INT1	55	PD11	
IMU_INT2	56	PD12	
IMU_CS	57	PD13	
TX4	60	PD14	
RX4	61	PD15	
OSC_IN	13	PH0-OSC_IN	
	8	PC15-OSC32_OUT	CH2_Fire
	14	PH1-OSC_OUT	OSC_OUT
	25	PA0	MES_BATT_1
	26	PA1	MES_BATT_2
	27	PA2	I_SENSE_5V
	28	PA3	I_SENSE_BAT
	31	PA4	LED_2
	32	PA5	CH6_PWM_MCU
	33	PA6	CH5_PWM_MCU
	34	PA7	CH4_PWM_MCU
	66	PA8	LED_1
	67	PA9	
	68	PA10	
	69	PA11	USB_N
	70	PA12	USB_P
	71	PA13	SWDIO
	77	PA14	SWCLK
	78	PA15	
	16	PC0	CAM_1
	17	PC1	CAM_2
	18	PC2_C	CAM_3
	19	PC3_C	
	35	PC4	
	36	PC5	
	62	PC6	GPS_RST
	63	PC7	GPS_INT
	64	PC8	DAT0_SD
	65	PC9	DAT1_SD
	79	PC10	DAT2_SD
	80	PC11	DAT3_SD
	81	PC12	CLK_SD
	6	PC13	Arming
	7	PC14-OSC32_IN	CH2_Cont
	1	PE2	CH1_Fire
	2	PE4	CH1_Cont
	3	PE5	Buzzer
	40	PE7	TX6
	41	PE8	RX6

U4B — 2 of 2 — STM32H725VGT6

VBAT	5	
VCAP	44	
VCAP	72	
VCAP	97	
VDD	4	
VDD	20	
VDD	30	
VDD	47	
VDD	59	
VDD	75	
VDD	88	
VDD	100	
VDD	46	
VDDLDO	74	
VDDLDO	99	
VDDLDO	76	
VDD33USB	24	
VDDA	11	
VDDSMPS		
VFBSMPS	12	
VLXSMPS	10	
VREF+	23	
VSS	21	
VSS	29	
VSS	45	
VSS	58	
VSS	73	
VSS	98	
VSSA	22	
VSSSMPS	9	

STM 620mA może wziąć max

R94 0603 10KΩ — MES_BATT_1 — D30 MM3Z3V3T1G — R95 4.7k 0603 — R96 0603 10KΩ — GND

BATT_IN_2 — R34 0603 10KΩ — MES_BATT_2 — D27 MM3Z4V7T1G — R38 4.7k 0603 — R46 0603 10KΩ — GND

HSE CRYSTAL

OSC_IN — Y1 — OSC_OUT; C34 6pF; C35 6pF; NC NC; XRCGE50M000F3A1AR0; GND

BOOT & NRST

R44 1KΩ 0603 — 3V3 — SW1 JS102011JCQN — BOOT — P1 — 62130621021 — NRST — C19 100nF 100V 0603 — SW2 PTS810SJM250SMTRLFS — R45 0603 10KΩ — GND

PROGRAMER

3V3 — P2 — SWCLK — SWDIO — 61300411121 — GND

Pinout zgodny z programatorem ST-Link V2 mini

PORTS

UART
TX6 RX6 — TX RX — UART6 — Main[2B]

I2C
SDA1 SCL1 — SDA SCL — I2C1 — Main[2B]

SPI
SPI2_CLK SPI2_SDI SPI2_SDO — SCLK SDI SDO — SPI2 — Main[2C]

USB
USB_N USB_P — USB_N USB_P — USB_PROG — Main[2B], USB[2B]

PYRO
CH1_Fire CH2_Fire Arming CH1_Cont CH2_Cont — Pyro_control

CAN
CAN_TX CAN_RX — CAN_TX CAN_RX — CAN — Main[3B], CAN[3C]

SDMMC
DAT2_SD DAT3_SD CMD_SD CLK_SD DAT0_SD DAT1_SD DETECT_SD — SDMMC — Main[3C], SD_CARD[4A]

GPS
TX4 RX4 GPS_RST GPS_INT — TX RX GPS_RST GPS_INT — GPS

INDICATORS
Buzzer LED_1 LED_2 — Indicators_control

BARO_CS MAG_CS MAG_INT IMU_CS IMU_INT1 IMU_INT2 — Sensors_control — Main[2C], Sensors[3C]

PWM_SERVO
CH1_PWM_MCU CH2_PWM_MCU CH3_PWM_MCU CH4_PWM_MCU CH5_PWM_MCU CH6_PWM_MCU — Servos_control

LORA
SPI6_CLK SPI6_MISO SPI6_MOSI DIO1 NSS RST DIO0 DIO3 — SCK MISO MOSI DIO1 NSS RST DIO0 DIO3 — LORA — Main[2C], LORA[4A]

CAMERA
CAM_1 CAM_2 CAM_3 — Camera_1 Camera_2 Camera_3 — CAMERA

CURR SENSE
I_SENSE_5V I_SENSE_BAT — I_SENSE_5V I_SENSE_BAT

Decoupling

VDD MCU FILTERING
3V3 — C41 C42 C43 C44 C45 C46 C47 C48 100nF 50V 0603; C40 4.7 uF 0402 16 V — GND

VDDSMPS
3V3 — C49 1 uF 0402 16 V; C50 4.7 uF 0402 16 V — GND

VDD33USB
3V3 — C52 100nF 50V 0603; C51 1 uF 0402 16 V — GND

VBAT & VDDA & VREF+VDD Decoupling
3V3 — C56 100nF 50V 0603; C59 1 uF 0402 16 V — C57 100nF 50V 0603; C60 1 uF 0402 16 V — C58 100nF 50V 0603; C53 1 uF 0402 16 V — GND

VLXSMPS
VLX — C61 220 pF 0402 50 V — GND

VFBSMPS
VFB — C16 10V 10µF 0402; C28 10V 10µF 0402 — GND

PULLUP

SDA1 — R42 4.7k 0603 — 3V3
SCL1 — R43 4.7k 0603 — 3V3

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	
REFERENCE DOCUMENTS	
BOM:	
ASSY DWG:	
FAB DWG:	
PCB DWG:	

PROJECT — Altium

PROJECT REVISION: 099ac0179365a291fce65 — DOCUMENT REVISION: — DESIGN ITEM: 386eadc3f5f9c224298

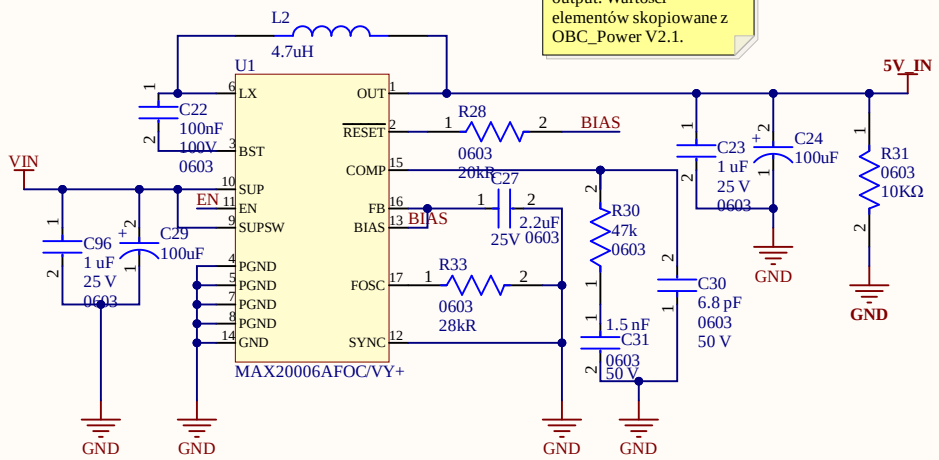
TITLE *

SIZE	CAGE CODE	DWG NO.	REV
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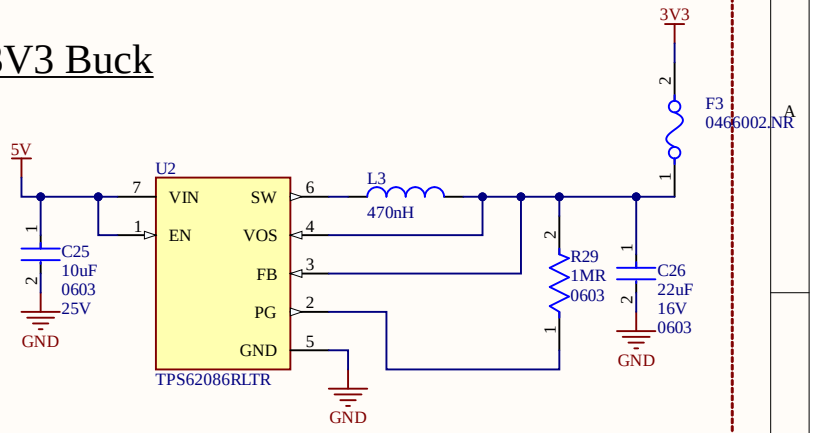
SCALE: — FILE NAME MCU.SchDoc — SHEET * OF *

7.4V TO 5V - BUCK CONVRTER

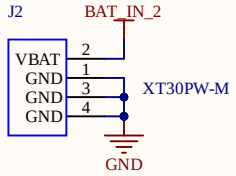
Vin pomiędzy 2S a 4S. 1Mhz częstotliwość. 5V output. Wartości elementów skopiowane z OBC_Power V2.1.



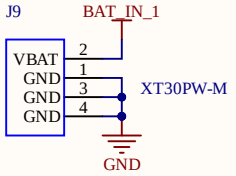
3V3 Buck



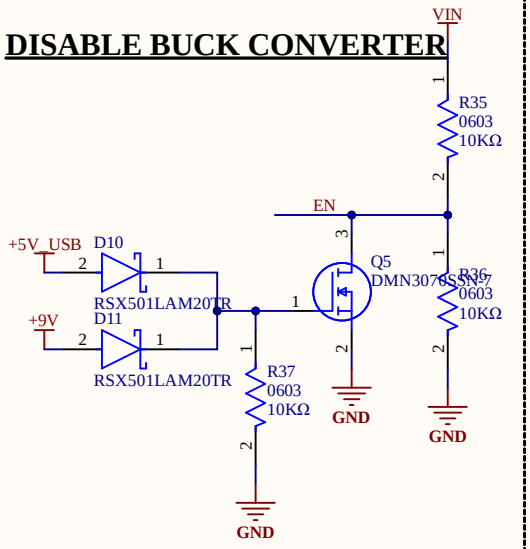
BATT CONNECTOR



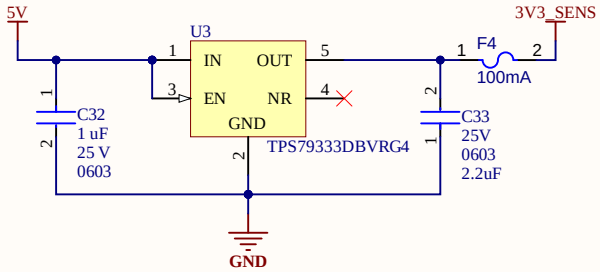
BATT CONNECTOR



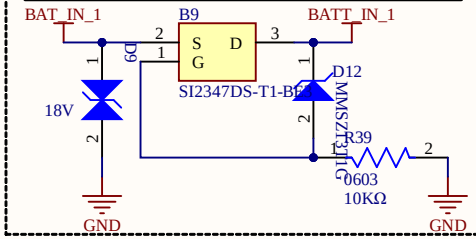
DISABLE BUCK CONVERTER



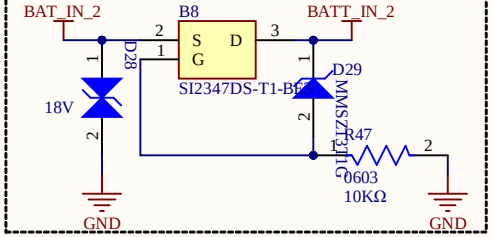
LDO 5V TO 3.3V for Sensors



REVERSE POLARITY

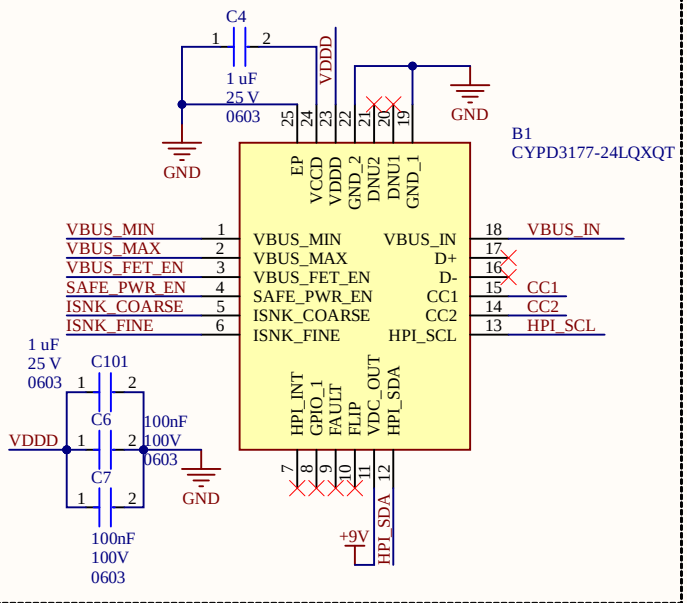


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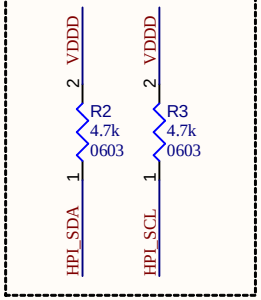


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	Sheet number: * / * ***Modified:*** 23.08.2026
Description: ***Reviewers:***	

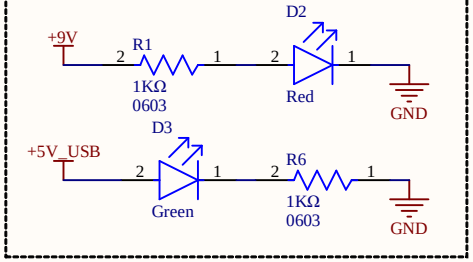
Power Delivery USB-C



Pull up I2C



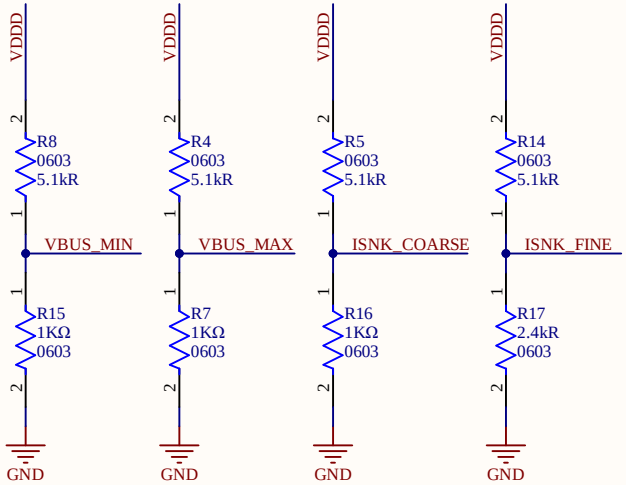
Charging status



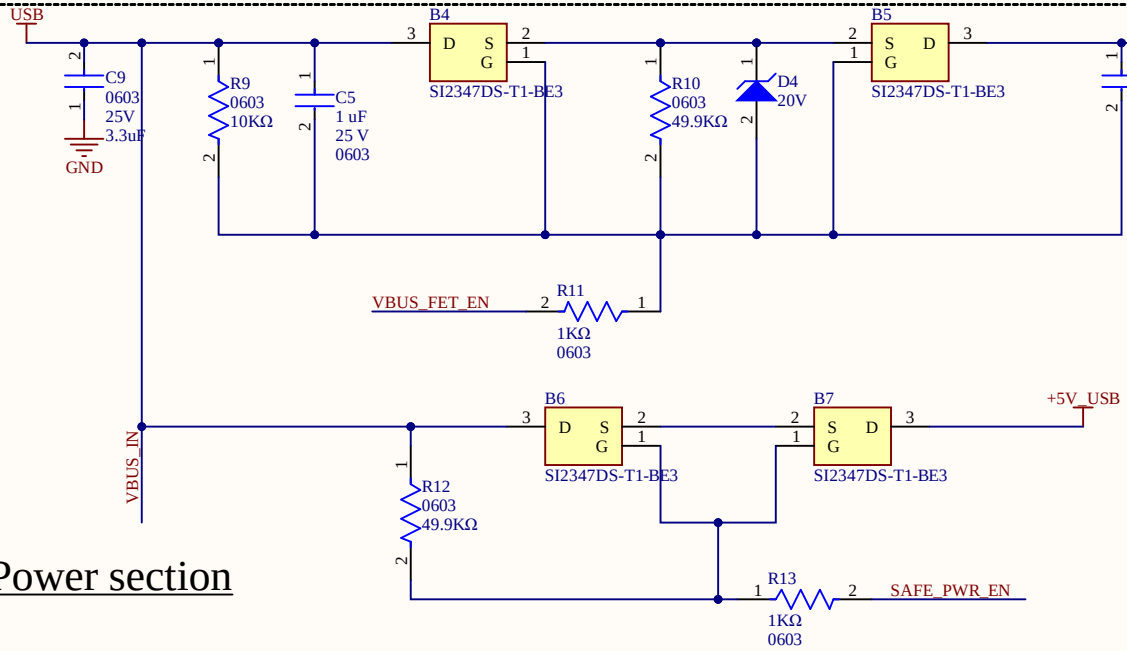
Ports



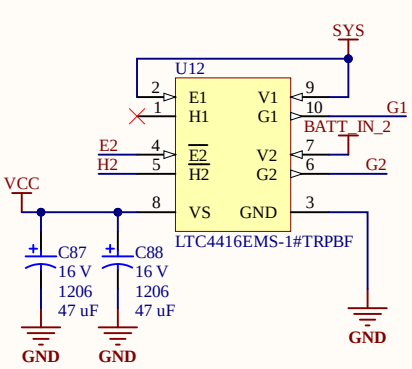
Voltage & Current determine

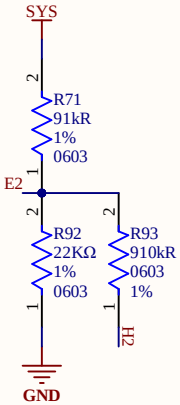


Power section

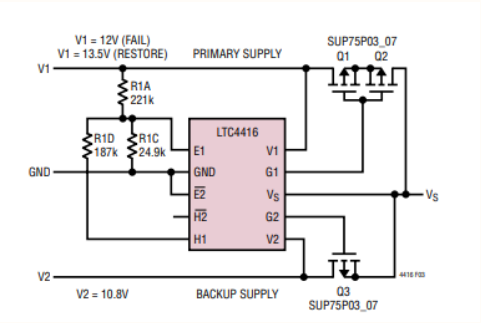


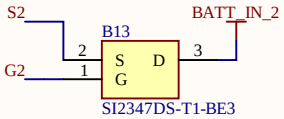
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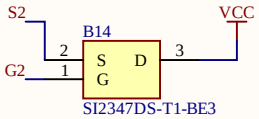


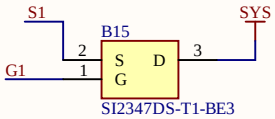


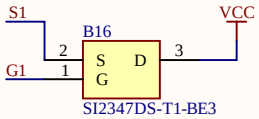
Dzielnik
Wartości
Vfail 6,28V
VRecover 6,4V
SYS priority





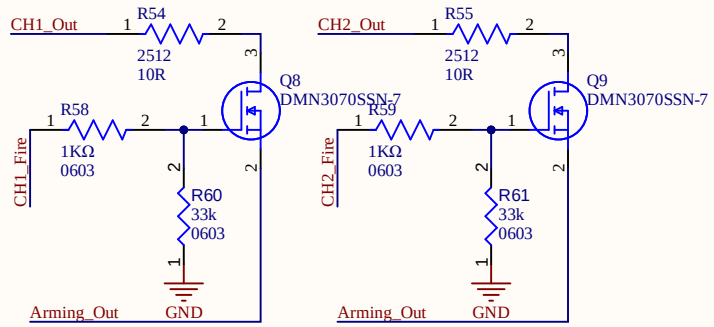




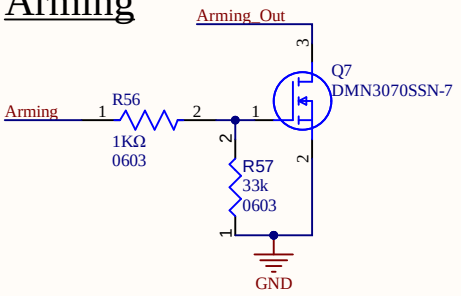


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Date: 9.23.2026		Sheet of
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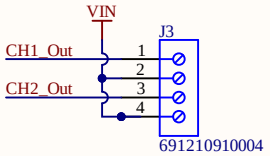
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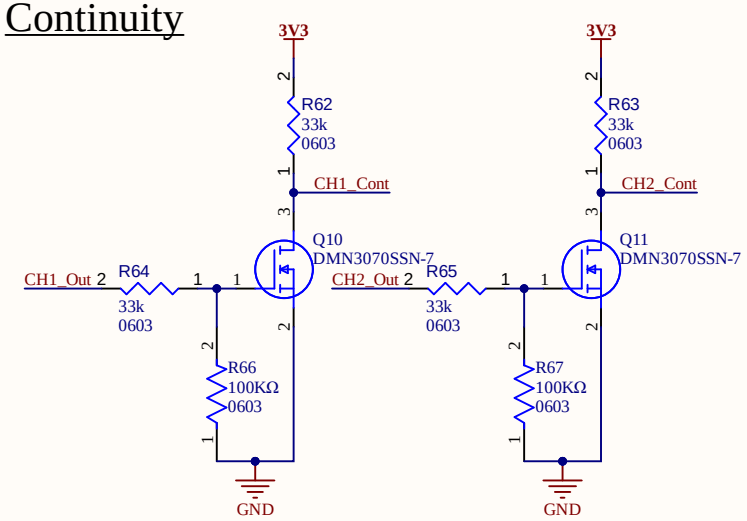
Arming



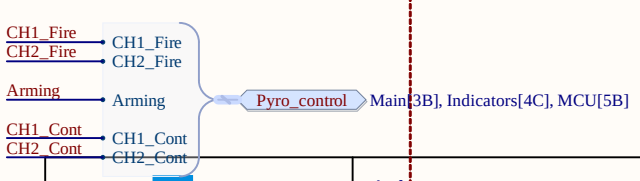
Output



Continuity



Ports



PWR IN SPACE

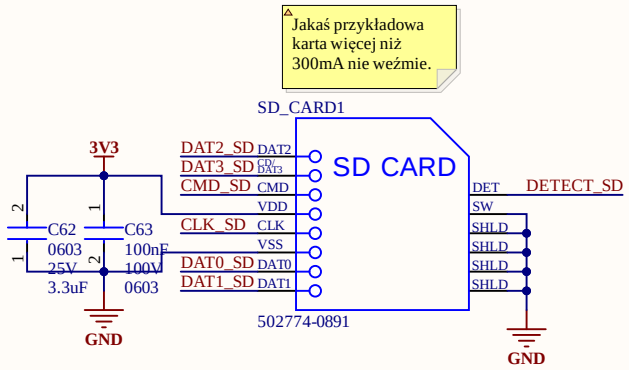
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Title:

Sheet number: * / *
Modified: 23.08.2026

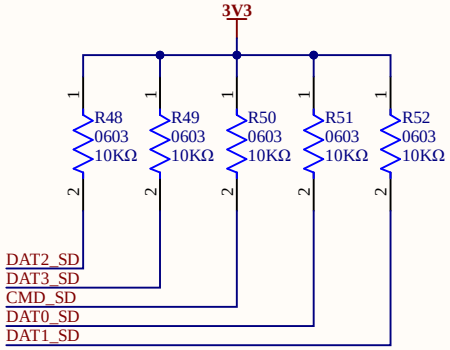
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Reviewers:

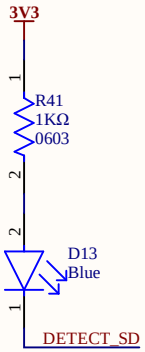
SD_CARD



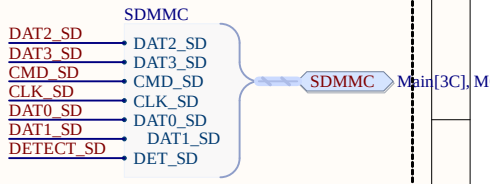
PULL UP



LED_STATUS



PORTS



PWR IN SPACE

Author:
Title:

Sheet number: * / *
Modified: 23.08.2026

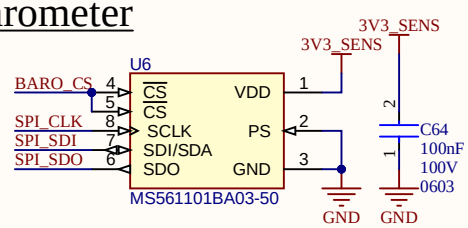
Description:

Reviewers:

Sensors

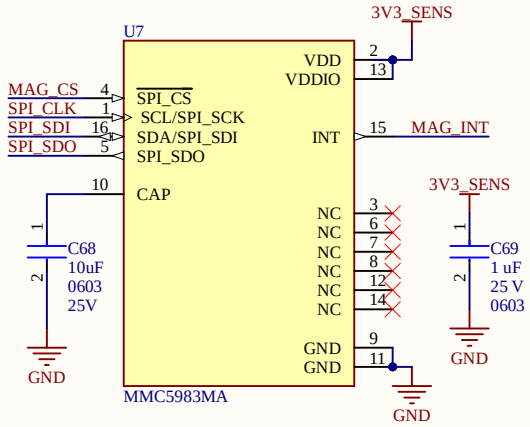
Imax = 1.4mA

Barometer



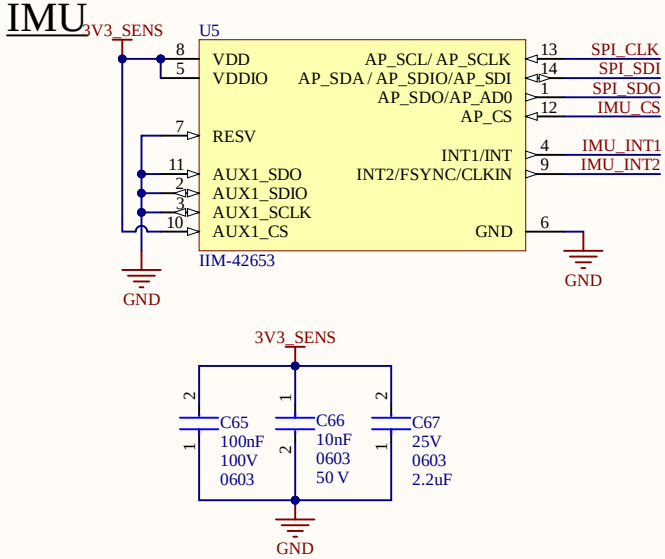
Imax = 32u / 7 * 1000 = 4.6mA

Magnetometer

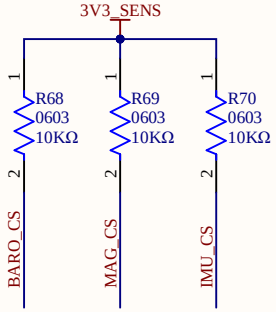


Imax = 0.88mA dla 1.8VDD. W takim razie zakładam 2x taką wartość. Czyli 1.76mA

IMU



PULLUP



SPI



PWR IN SPACE

Author:
Title:

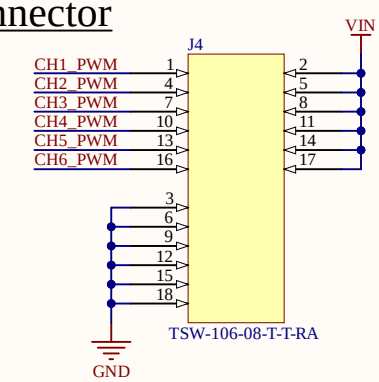
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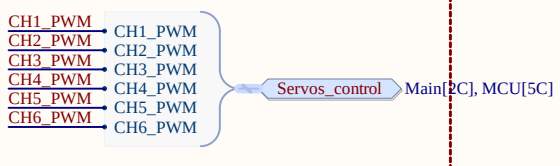
Reviewers:

Servos

Connector



Ports



PWR IN SPACE

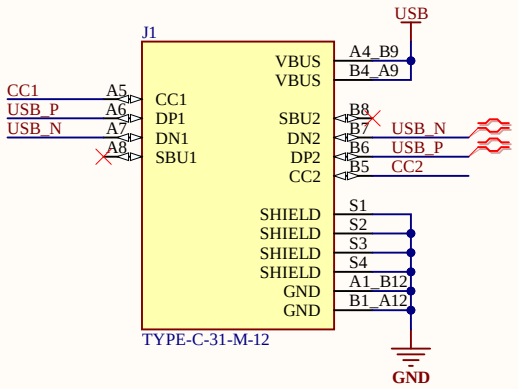
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Title:

Sheet number: * / *
Modified: 19.08.2026

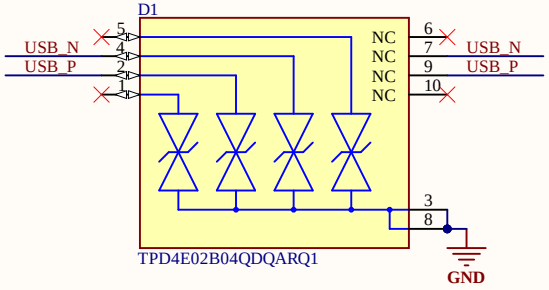
Description:

Reviewers:

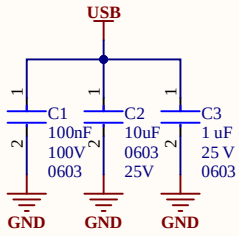
STM USB TO PROG (USB C 2.0)



OVERVOLT PROTECT



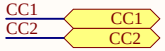
USB 5V FILTER

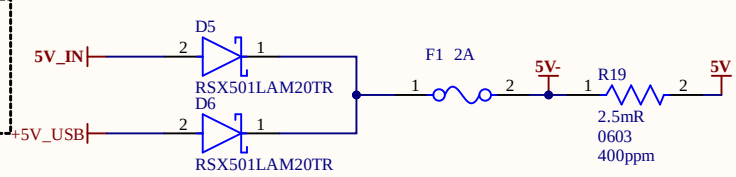


USB



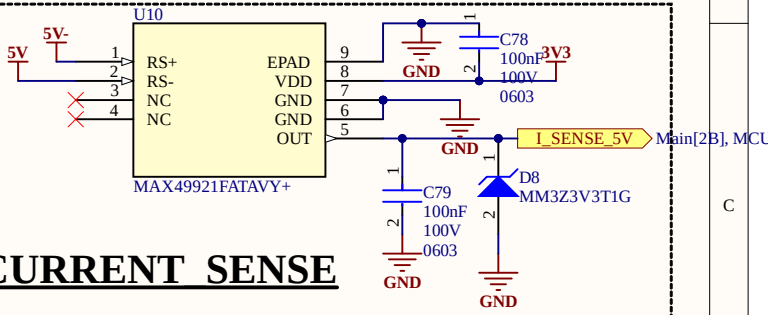
Ports





MASS

CURRENT SENSE



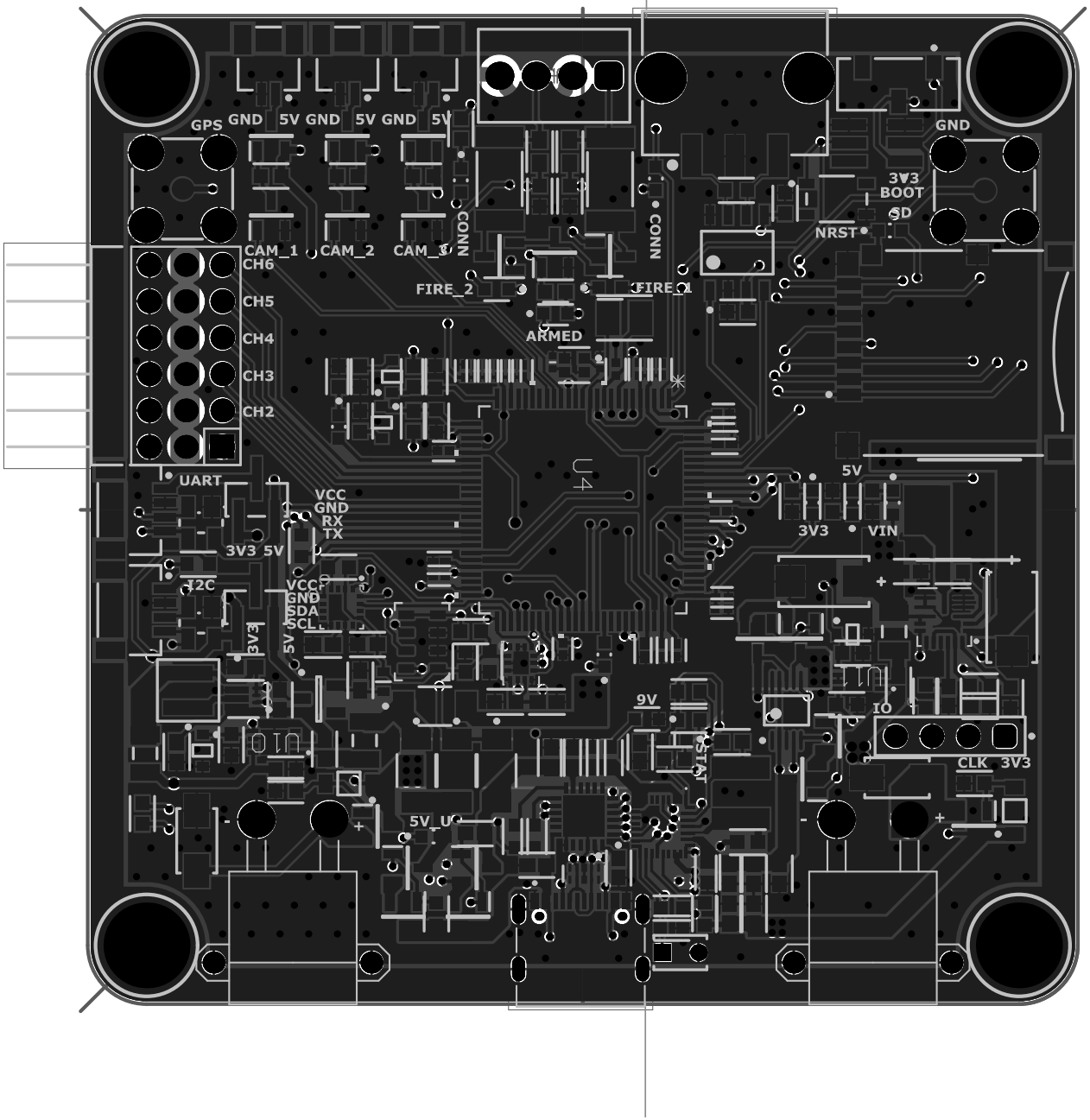
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Title:

Sheet number: * / *
Modified: 24.08.2026

Description:

Reviewers:


GPS GND 5V GND 5V GND 5V
GND
3V3
BOOT
SD
NRST
CONN
CONN
CAM_1 CAM_2 CAM_3
CH6
CH5
CH4
CH3
CH2
FIRE_2
FIRE_1
ARMED
U4
UART
VCC
GND
RX
TX
3V3 5V
I2C
VCC
GND
SDA
SCL
3V3
5V
5V
3V3
VIN
9V
IO
CLK 3V3
5V_U
STLK

Board Stack Report